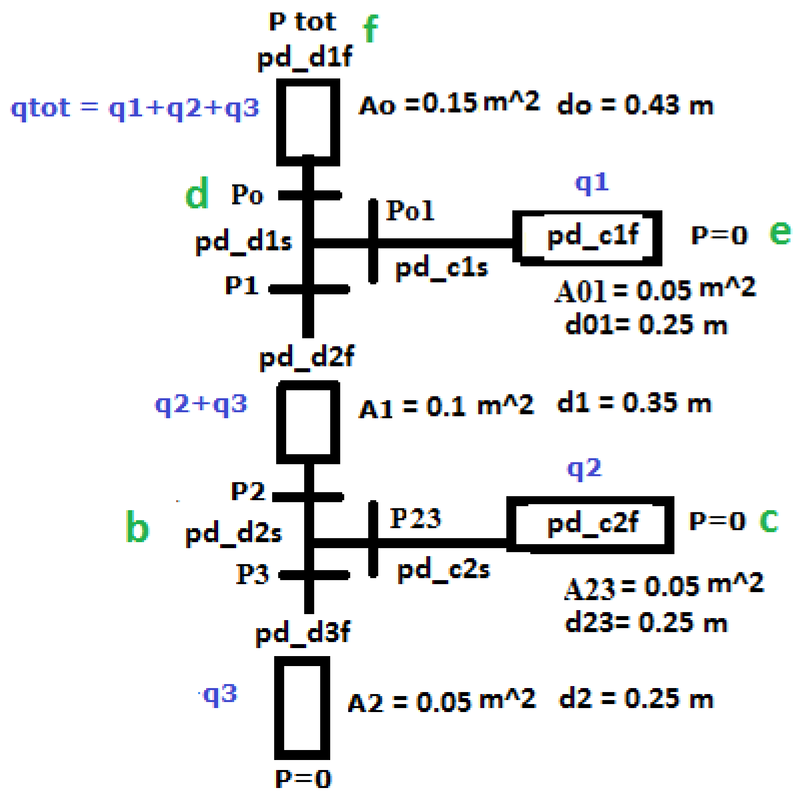

Friction losses (Darcy Weisbach)

$$\Delta p = f_D \cdot \frac{L}{D} \cdot \frac{\rho V^2}{2}$$

where the pressure loss due to friction Δp (Pa) is a function of:

- the ratio of the length to diameter of the pipe, L/D;
- the density of the fluid, ρ (kg/m^3);
- the mean velocity of the flow, V (m/s), as defined above;
- Darcy Friction Factor; a (dimensionless) coefficient

Where the Swamee–Jain equation is used to solve directly for the Darcy–Weisbach friction factor *f* for a full-flowing circular pipe. It is an approximation of the implicit Colebrook–White equation.

$$f = \frac{0.25}{\left[\log_{10}\left(\frac{\varepsilon}{3.7D} + \frac{5.74}{\mathrm{Re}^{0.9}}\right)\right]^2}$$

Where

$D =$ diameter (m)
$\rho =$ air density (1.204 kg/m$^{3)}$)
V= fluid mean velocity $u = {}^{q}/_{A}$ (m/s)
ε= Roughness height (0.15 mm for thin plate ducts)
Re= Reynolds number$Re = {}^{u \cdot d}/_{v}$
v = Kinematic viscosity (Air=15.1 x 10^{-6} m^2/s)

Single losses

$$p_\Delta = \zeta \cdot \frac{\rho \cdot u^2}{2}$$

ζ the resistance coefficient ζ is determined according to tables obtained from Danvak 'Climatic systems', 2. ed. 1997 eds. H.E. Hansen P. Kjerulf Jensen, Ole B. Stampe.

Distribution ducts:

$$P_{tot} - P_0 = p_{d\_d1\,f} \cdot \left(\frac{q1 + q2 + q3}{A0}\right)^2$$

$$P_0 - P_1 = p_{d\_\,d1s} \cdot \left(\frac{q2 + q3}{A1}\right)^2 = \zeta \cdot \frac{\rho}{2}$$

$$P_1 - P_2 = p_{d\_d2f} \cdot \left(\frac{q2 + q3}{A1}\right)^2$$

$$P_2 - P_3 = p_{d\_\,d2s} \cdot \left(\frac{q3}{A2}\right)^2$$

$$P_3 = p_{d\_\,d3f} \cdot \left(\frac{q3}{A2}\right)^2$$

Connection ducts:

$$P_0 - P_{01} = p_{d\_\,c1s} \cdot \left(\frac{q1}{A01}\right)^2$$

$$P_{01} - 0 = p_{d\_\,c1\,f} \cdot \left(\frac{q1}{A01}\right)^2$$

$$P_2 - P_{23} = p_{d\_\,c2s} \cdot \left(\frac{q2}{A23}\right)^2$$

$$P_{23} - 0 = p_{d\_\,c2\,f} \cdot \left(\frac{q2}{A23}\right)^2$$

Where

Duct friction resistance (includes the unknowns, calculated in each time step based on q act)

$$p_{d\_f} = f_D \cdot \frac{L \cdot \rho}{2 \cdot d} = \frac{0.25}{\left[\log_{10}\left[\frac{5.74}{Re^{0.9}} + \frac{\varepsilon}{3.7\, d}\right]\right]^2} \cdot \frac{L \cdot \rho}{2 \cdot d}$$

Assign
c1 = L * 1.204/(2*d)
c2 = 0.00015/(3.7*d)

$$p_{d\_f} = \frac{0.25}{\left[\log_{10}\left[\frac{5.74}{\left(\frac{d\,\frac{q}{A}}{\nu}\right)^{0.9}} + c2\right]\right]^2} \cdot c1$$

Branch dynamic resistance (does not include the unknowns, calculated in each time step based on q set, since it is difficult to obtain the ζ coefficient for the unknown flow rates)

$$p_{d\_s} = \zeta \cdot \frac{\rho}{2}$$

The vector of unknowns is

$$\boldsymbol{x} = (Po, P1, P2, P3, P01, P23, q1, q2, q3)$$

The system of non-linear algebraic equations can be put in the following form

$$\boldsymbol{F}(\boldsymbol{x}) = \boldsymbol{F}(Po, P1, P2, P3, P01, P23, q1, q2, q3) = 0$$

Where

$$\boldsymbol{F} = (F1, F1, F3, F4, F5, F6, F7, F8, F9)$$

Newton's Method for systems can be applied

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n + \boldsymbol{F}(\boldsymbol{x}_n)/\boldsymbol{F}'(\boldsymbol{x}_n)$$

With

$$F1 = P_{tot} - P_0 - p_{dd1f} \cdot \left(\frac{q1 + q2 + q3}{A_o}\right)^2 = 0$$

$$F2 = P_0 - P_1 - p_{dd1d} \cdot \left(\frac{q2 + q3}{A_1}\right)^2 = 0$$

$$F3 = P_1 - P_2 - p_{dd2f} \cdot \left(\frac{q2 + q3}{A_1}\right)^2 = 0$$

$$F4 = P_2 - P_3 - p_{dd2d} \cdot \left(\frac{q3}{A_2}\right)^2 = 0$$

$$F5 = P_3 - p_{dd3f} \left(\frac{q3}{A_2}\right)^2 = 0$$

$$F6 = P_0 - P_{01} - p_{dc1d} \cdot \left(\frac{q1}{A_{01}}\right)^2 = 0$$

$$F7 = P_{01} - p_{dc1f} \cdot \left(\frac{q1}{A_{01}}\right)^2 = 0$$

$$F8 = P_2 - P_{23} - p_{dc2d} \cdot \left(\frac{q2}{A_{23}}\right)^2 = 0$$

$$F9 = P_{23} - p_{dc2f} \cdot \left(\frac{q2}{A_{23}}\right)^2 = 0$$

Constrain q1 + q2 +q3 = q

Constrain q1, q2, q3 $\geq$ 0.030

Constrain q1, q2, q3 $\leq$ 0.080

The corresponding set of equations in Matlab is given below:

```
[Pt_act-x(1)-((0.25/(log10(c02+5.74/(((x(7)+x(8)+x(9))/A0)*d0/0.0000151)^0.9))^2)*c01*((x(7)+x(8)+x(9))/A0)^2);
x(1)-x(2)-pd_d1s*((x(8)+x(9))/A1)^2;
x(2)-x(3)-((0.25/(log10(c12+5.74/(((x(8)+x(9))/A1)*d1/0.0000151)^0.9))^2)*c11*((x(8)+x(9))/A1)^2);
x(3)-x(4)-pd_d2s*(x(9)/A2)^2;
x(4)-((0.25/(log10(c22+5.74/((x(9)/A2)*d2/0.0000151)^0.9))^2)*c21*(x(9)/A2)^2);
x(1)-x(5)-pd_c1s*(x(7)/A01)^2;
x(5)-((0.25/(log10(c012+5.74/((x(7)/A01)*d01/0.0000151)^0.9))^2)*c011*(x(7)/A01)^2);
x(3)-x(6)-pd_c2s*(x(8)/A23)^2;
x(6)-((0.25/(log10(c232+5.74/((x(8)/A23)*d23/0.0000151)^0.9))^2)*c231*(x(8)/A23)^2);
x(7)+x(8)+x(9)-q];
```